

Class Checking: `instanceof`

`instanceof` kya karta hai?

Check karta hai ki koi object kisi class ka hai ya nahi. **Inheritance bhi count hoti hai.**

javascript

```
obj instanceof Class // true / false
```

javascript

```
class Animal {}  
class Rabbit extends Animal {}  
let rabbit = new Rabbit();  
  
rabbit instanceof Rabbit; // true  
rabbit instanceof Animal; // true (parent bhi)  
  
[1, 2, 3] instanceof Array; // true  
[1, 2, 3] instanceof Object; // true
```

Constructor functions ke saath bhi chalta hai.

Andar se kaise chalta hai?

1. Agar class me static method `Symbol.hasInstance` ho, to wahi call hota hai aur uska `true/false` final hota hai (custom logic ke liye).
2. Warna JS object ki **prototype chain** me `Class.prototype` dhoondhta hai:

javascript

```
obj.__proto__ === Class.prototype ?  
obj.__proto__.__proto__ === Class.prototype ?  
... (chain ke end tak)
```

Kahin match mila to `true`, warna `false`.

Ek aur tareeka: `Class.prototype.isPrototypeOf(obj)` (same kaam).

Ajeeb baat: check me **constructor function khud شامل nahi hota**, sirf `Class.prototype` aur chain dekhi jaati hai. Isliye agar object banne ke baad `Rabbit.prototype` badal do, to `rabbit instanceof Rabbit` `false` ho jaata hai.

Bonus: `Object.prototype.toString` se type

Ye `typeof` ka powerful version hai. Isse call karo `.call(value)` ke saath:

javascript

```
let s = Object.prototype.toString;

s.call([]);           // [object Array]
s.call(123);          // [object Number]
s.call(null);         // [object Null]
s.call(alert);        // [object Function]
```

`Symbol.toStringTag`

Is result ko customize kar sakte ho:

javascript

```
let user = { [Symbol.toStringTag]: "User" };
{}.toString.call(user); // [object User]
```

Browser ke objects me (`window`), (`XMLHttpRequest`) ye pehle se hota hai.

Type checking ke tareeke

Tareeka	Kis par chalta hai	Deta kya hai
<code>typeof</code>	primitives	string
<code>{}.toString</code>	primitives, built-in objects, <code>toStringTag</code> wale	string
<code>instanceof</code>	objects	true/false

Task: "Strange instanceof"

javascript

```
function A() {}
function B() {}
A.prototype = B.prototype = {};
let a = new A();
a instanceof B; // true
```

Kyunki `instanceof` sirf **prototype** dekhta hai, function nahi. `a.__proto__ == B.prototype`, isliye `true`.

Yaad rakho

- `instanceof` = class hierarchy check karne ke liye best.
- Built-in type string chahiye to `{}.toString.call(x)`.
- Type asal me prototype se decide hota hai, constructor se nahi.